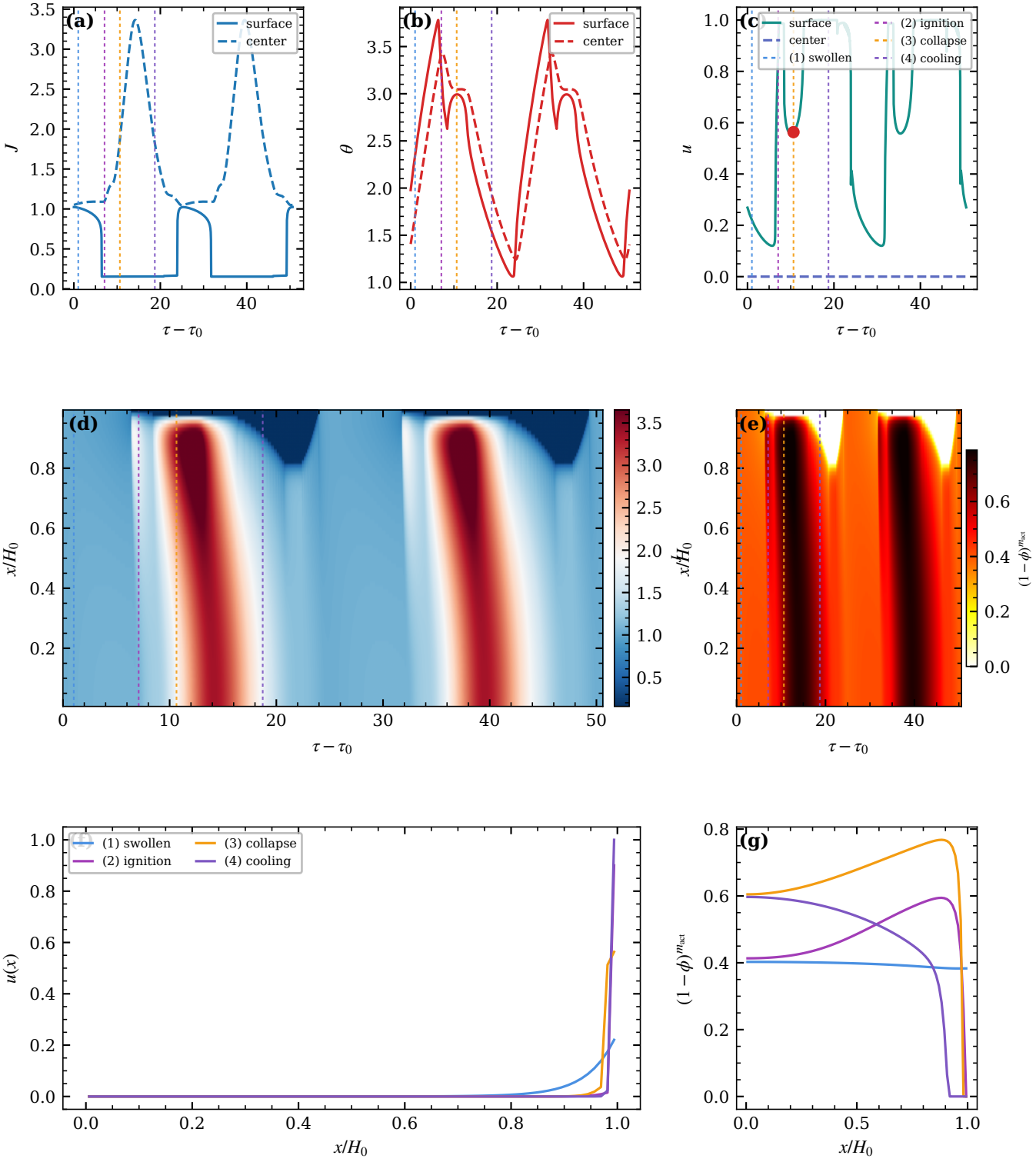
(a)
surface
center
J
$\tau - \tau_0$
(b)
surface
center
θ
$\tau - \tau_0$
(c)
surface
center
(1) swollen
(2) ignition
(3) collapse
(4) cooling
u
$\tau - \tau_0$
(d)
x/H_0
$\tau - \tau_0$
(e)
x/H_0
$(1-\phi)^{n_{act}}$
$\tau - \tau_0$
(1) swollen
(2) ignition
(3) collapse
(4) cooling
$u(x)$
x/H_0
(g)
$(1-\phi)^{n_{act}}$
x/H_0